CHEMISTRY ADAPTLY

# GROUP 7 – THE HALOGENS

## KEY VOCABULARY

**Halogen:** a Group 7 non-metal with seven outer electrons.

**Diatomic:** exists as molecules of two atoms, e.g. $Cl_2$.

**Displacement:** a more reactive halogen pushes out a less reactive one.

**Ion formed:** halogens gain 1 electron to form a 1- ion (halide).

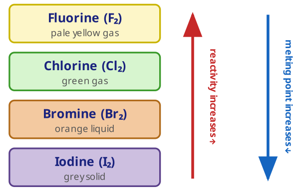
## COMMON MISCONCEPTIONS

~~Reactivity increases down Group 7.~~
✓ Reactivity decreases down the group.

~~Halogens form 1+ ions.~~
✓ They gain an electron to form 1- ions.

~~All halogens are gases.~~
✓ States change: gas → liquid → solid down the group.

## TRENDS DOWN GROUP 7




## 1 OUTER ELECTRONS

State the number of **outer electrons** in a halogen atom. **[1 mark]**

## 2 REACTIVITY TREND

**Explain** why reactivity decreases down Group 7. **[3 marks]**

*Think about how easily the outer shell gains an electron.*

## 3 DISPLACEMENT

Chlorine is added to potassium bromide. **Predict** what happens and why. **[2 marks]**

## 4 FORM A HALIDE ION

Describe how a chlorine atom forms a **chloride ion**. **[1 mark]**

## 5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

*A halogen _____ one electron. Reactivity _____ down the group.*

gains | decreases | loses | increases

## 6 WRITE AN EQUATION

Write the equation for chlorine displacing bromine from KBr. **[2 marks]**

***TIP:*** *Smaller atom → outer shell closer → gains an electron more easily → more reactive.*

**✓ Think: Why can fluorine displace every other halogen?**